

LAURENS COUNTY AP, SC, USA

WMO: 720597

Lat: 34.500N

Lon: 81.950W

Elev: 696

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00190

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	27.1	5.4	7.4	28.5	10.2	9.5	32.1	19.7	57.2	17.8	54.1	4.0	20	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	94.8	76.0	91.3	74.9	90.3	74.6	81.2	87.4	79.3	85.0	77.5	84.4	6.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	156.2	84.6	77.4	146.3	82.2	75.5	136.9	80.3	45.4	88.6	43.6	85.2	41.2	83.3	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.5	11.7	14.7	97.0	8.7	2.2	8.5	98.5	3.4	99.8	-1.5	101.0	-7.8	102.6
			12.6	80.3	7.9	2.4	6.9	82.1	2.3	83.5	-2.2	84.8	-7.9	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	42.7	47.2	52.8	62.0	70.1	77.1	79.5	78.1	73.5	62.1	51.4	46.4
	DBStd	14.81	9.39	9.41	9.01	7.15	6.19	4.28	3.55	3.62	5.34	7.58	7.32	8.87
	HDD50	753	263	154	76	5	0	0	0	0	0	5	69	181
	HDD65	2890	691	503	387	140	31	0	0	0	6	144	409	579
	CDD50	5131	37	74	164	365	622	813	916	873	705	380	112	70
	CDD65	1793	0	4	10	50	188	363	451	408	261	54	2	2
	CDH74	15950	2	12	108	495	1551	3355	4319	3537	2108	450	12	1
	CDH80	6436	0	0	9	85	488	1435	1981	1501	831	106	0	0
Wind	WSAvg	4.9	5.4	6.1	6.1	6.1	4.9	4.2	4.2	3.6	3.9	4.6	4.6	5.0
Precipitation	PrecAvg	47.6	4.6	4.2	5.2	3.6	3.6	4.1	3.5	4.1	3.8	3.4	3.5	3.8
	PrecMax	67.6	7.8	8.1	10.9	10.5	10.2	12.7	10.2	10.8	9.2	14.2	10.4	8.7
	PrecMin	32.4	0.6	0.6	0.9	0.6	0.6	0.3	0.6	0.1	0.0	0.0	0.5	0.7
	PrecStd	7.2	1.7	1.8	2.3	2.1	2.1	2.5	2.0	2.5	2.2	2.8	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	75.4	81.4	86.0	91.3	95.3	97.1	95.3	93.2	89.5	75.2	72.3
		MCWB	61.5	62.8	62.3	67.0	70.2	77.8	77.3	75.5	73.3	73.1	64.3	63.9
	2%	DB	64.5	71.5	77.3	82.0	88.3	92.9	95.0	91.4	90.6	81.9	71.9	68.5
		MCWB	57.9	61.0	61.0	64.9	70.0	75.8	76.8	74.6	72.8	68.3	60.8	63.5
	5%	DB	62.7	67.7	72.7	80.6	85.9	90.4	91.3	90.4	88.3	79.1	68.3	64.3
		MCWB	56.7	57.7	59.0	64.7	70.3	74.9	76.1	74.4	72.3	66.5	59.7	60.2
	10%	DB	58.6	63.2	69.7	76.9	81.8	88.2	90.1	88.2	85.8	75.1	64.2	61.5
		MCWB	52.0	56.0	57.7	62.9	69.4	74.1	75.9	74.3	71.8	65.5	57.6	57.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	65.7	66.1	72.2	76.4	82.3	83.1	83.2	78.0	75.1	68.1	68.5
		MCDB	66.3	72.7	74.2	79.6	83.3	91.7	92.2	89.2	84.6	81.6	69.9	69.7
	2%	WB	61.7	63.4	64.1	69.4	74.3	80.7	81.2	80.9	76.2	72.9	64.7	64.5
		MCDB	64.6	67.0	71.1	75.2	81.9	88.6	87.7	84.2	84.6	78.9	68.9	66.6
	5%	WB	57.6	60.5	62.4	67.9	72.6	78.6	79.2	78.3	74.5	70.8	63.0	61.4
		MCDB	61.9	65.4	69.2	75.3	80.2	84.9	85.5	83.2	83.1	75.2	65.4	63.1
	10%	WB	53.2	56.9	60.2	65.8	71.3	76.2	77.3	76.7	73.2	67.9	59.9	57.4
		MCDB	56.5	62.0	66.5	74.0	78.8	83.4	85.4	83.6	81.5	73.2	63.3	61.6
Mean Daily Temperature Range	5% DB	MDBR	19.1	19.6	21.9	23.2	20.7	20.0	19.6	18.9	19.9	21.9	21.4	17.6
		MCDBR	22.9	23.4	25.8	25.3	24.5	22.9	22.4	22.0	24.4	24.3	24.8	19.0
		MCWBR	15.8	14.2	13.0	11.0	9.7	8.4	7.4	7.6	9.1	12.1	15.3	13.4
	5% WB	MCDBR	19.2	19.2	21.5	20.8	19.7	20.4	20.5	18.0	21.0	19.5	18.0	15.4
		MCWBR	16.2	13.3	12.9	11.2	9.9	10.0	9.3	9.0	8.7	10.9	13.6	12.8
Clear-Sky Solar Irradiance	taub	0.315	0.325	0.361	0.401	0.459	0.487	0.535	0.520	0.429	0.390	0.341	0.315	
	taud	2.501	2.474	2.399	2.279	2.158	2.121	2.011	2.040	2.274	2.360	2.457	2.532	
	Ebn at Noon	284	293	289	281	265	256	243	244	264	266	271	277	
	Edh at Noon	27	30	36	42	48	50	56	53	40	34	28	24	
All-Sky Solar Radiation	RadAvg	857	1057	1400	1754	1940	2048	1924	1764	1513	1255	949	738	
	RadStd	46	121	99	131	154	147	122	85	133	149	82	72	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+311	+211

Nomenclature: See separate page